



Barker
College

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Student Number

2021

YEAR 12 ASSESSMENT BLOCK
SEMESTER 1

Mathematics Extension 2

Section I - Multiple Choice

Write your student number at the top of this page.

Remove this sheet.

Select the alternative A, B C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ A 2 B 6 C 8 D 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

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Start
Here →

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

6. A ☐ B ☐ C ☐ D ☐

7. A ☐ B ☐ C ☐ D ☐

8. A ☐ B ☐ C ☐ D ☐



Barker
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Student Number

2021

**YEAR 12
ASSESSMENT BLOCK**

Mathematics Extension 2

Staff Involved:

BHC* ARM

12:00 NOON

FRIDAY 26 MARCH

40 copies

TOPICS COVERED:

Complex Numbers I & II

Proof

General Instructions

- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale

**Total marks:
80**

Section I – 8 marks (pages 4–7)

- Attempt Questions 1–8
- Allow about 15 minutes for this section

Section II – 72 marks (pages 8–13)

- Attempt Questions 9–14
- Allow about 1 hour and 45 minutes for this section

Section I

8 marks

Attempt Questions 1–8

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–8.

1 What is the negation of the following statement?

“If all rich people are happy, then all poor people are sad.”

- A. Not all rich people are happy and not all poor people are sad.
- B. All poor people are sad, but there is a rich person who is not happy.
- C. All rich people are happy, but there is a poor person who is not sad.
- D. Some rich people are not happy or some poor people are not sad.

2 If $|z - 3 - 3i| = 3$, what is the least value of $|z|$?

- A. $3\sqrt{2} - 3$
- B. $3\sqrt{2} - 2$
- C. $2\sqrt{3} - 2$
- D. $2\sqrt{3} - 3$

3 Which one of the following statements is true?

- A. $(\forall x, y \in \mathbb{R}) \left(\frac{x}{y} \in \mathbb{R} \right)$
- B. $(\exists x \in \mathbb{R})(x^2 + 4x + 5 = 0)$
- C. $(\forall x \in \mathbb{R})(x^2 \geq x)$
- D. $(\forall a \in \mathbb{Z}^+)(\exists x \in \text{primes})(a \leq x \leq 2a)$

4 Given that $z = \frac{\sqrt{3} + i}{1 - i}$, what is the smallest positive value of n for which z^n is purely imaginary?

- A. $\frac{6}{7}$
- B. $\frac{6}{5}$
- C. 6
- D. 12

5 When the cube roots of $2 - 2i$ are expressed in the form $re^{i\theta}$, $r = \sqrt{2}$. Which expression will give the correct values of θ ?

- A. $\frac{(8k-1)\pi}{4}$ for $k = -1, 0, 1$
- B. $\frac{(8k+1)\pi}{4}$ for $k = -1, 0, 1$
- C. $\frac{(8k+1)\pi}{12}$ for $k = -1, 0, 1$
- D. $\frac{(8k-1)\pi}{12}$ for $k = -1, 0, 1$

6 If $z = e^{3i\theta}$, which of the following expressions is equal to $1 + z^2$?

- A. $2 \cos^2 3\theta \operatorname{cis} 3\theta$
- B. $2 \cos 3\theta \operatorname{cis} 3\theta$
- C. $\cos^2 3\theta \operatorname{cis} 3\theta$
- D. $\cos 3\theta \operatorname{cis} 3\theta$

7 What is the contrapositive of the following statement?

“If Harry studies Biology or Chemistry in 2022,
then Jesse will not take a holiday in Fiji in 2022.”

- A. If Jesse does not take a holiday in Fiji in 2022,
then Harry will not study Biology or Chemistry in 2022.
- B. If Jesse takes a holiday in Fiji in 2022,
then Harry will study neither Biology nor Chemistry in 2022.
- C. If Jesse does not take a holiday in Fiji in 2022,
then Harry will study neither Biology nor Chemistry in 2022.
- D. If Jesse takes a holiday in Fiji in 2022,
then Harry will not study Biology or not study Chemistry in 2022.

8 Shanelle correctly wrote:

$$(\cos \theta)^3 = \left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right)^3$$

$$\therefore \cos^3 \theta = \frac{1}{8} \left(e^{3i\theta} + e^{-3i\theta} + 3(e^{i\theta} + e^{-i\theta}) \right)$$

From her last line of working, Shanelle should be able to deduce which one of the following?

- A. $\frac{3}{4} \cos \theta = \cos^3 \theta - \frac{1}{4} \cos 3\theta$
- B. $8 \cos^3 \theta = \cos 3\theta + 3 \cos \theta$
- C. $8 \cos^3 \theta = 2 \cos 3\theta + 3 \cos \theta$
- D. $\cos 3\theta = 4 \cos^3 \theta + 3 \cos \theta$

End of Section I

Section II

72 marks

Attempt Questions 9–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) $x - 2 - 2i$ is a factor of $P(x) = x^3 - ax^2 + 20x - 24$, where a is a real number.

Find the value of a and express $P(x)$ as a product of complex linear factors.

3

- (b) (i) Sketch the region in the complex plane where the following inequalities hold simultaneously:

$$|z - i| \leq 2 \quad \text{and} \quad 0 \leq \text{Arg}(z - 1) \leq \frac{3\pi}{4}$$

3

- (ii) What is the maximum value of $\text{Arg}(z)$?

Give your answer in radians, correct to 2 decimal places.

4

- (iii) By considering z as defined in the region above, what is the value of $\text{Arg}(|z|)$?

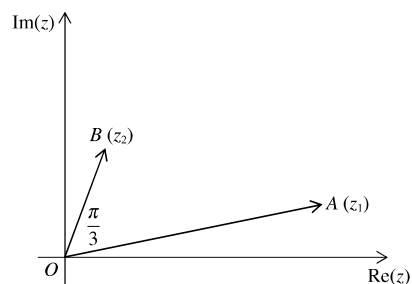
Give reasons for your answer.

2

Question 10 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) In the Argand diagram below, the point A represents the complex number z_1 and the point B represents the complex number z_2 . Furthermore, $\angle AOB = \frac{\pi}{3}$ and $|z_1| = 2 \times |z_2|$.



- (i) Express z_2 in terms of z_1 . Give your answer in exponential form. 1
- (ii) If $AOBC$ is a parallelogram, express the complex number represented by point C in terms of z_1 . Give your answer in exponential form. 1
- (iii) Express the complex number represented by the midpoint of BC in terms of z_1 . Give your answer in exponential form. 1
- (b) Prove by contradiction that there are no integers m and n which satisfy $3m + 21n = 136$. 2
- (c) Suppose that x and y are integers. Prove by contraposition that if $x^2(y + 3)$ is even, then x is even or y is odd. 3
- (d) (i) Starting with $(\sqrt{a} - \sqrt{b})^2 \geq 0$, show that $\frac{a+b}{2} \geq \sqrt{ab}$. 1
- (ii) Using the result in (i), prove that $xy + yz + zx \geq x\sqrt{yz} + y\sqrt{zx} + z\sqrt{xy}$, where $x, y, z \geq 0$. 3

Question 11 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) It is given that:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Using the identities above, prove that $e^{i\theta} = \cos \theta + i \sin \theta$.

3

- (b) Sketch the following locus on a complex plane: $\text{Arg} \left(\frac{z-1-i}{z+1-i} \right) = 0$. 3
- (c) Solve $z^4 - 8\sqrt{2} + 8\sqrt{2}i = 0$, leaving your answers in exponential form. 3
- (d) It is given that $|z| < \frac{1}{2}$.
Use a triangle inequality, or otherwise, to show that $|(1+i)z^3 + iz| < \frac{3}{4}$. 3

Question 12 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Given that $z = \text{cis}\left(\frac{2\pi}{3}\right)$ and $w = \sqrt{3} \text{cis}\left(\frac{\pi}{3}\right)$,
- (i) Express $\frac{w}{z}$ in modulus-argument form. **1**
- (ii) Use mathematical induction to prove that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for all integers $n \geq 0$. **3**
- (iii) Hence, by using the result in (ii), find the value of $\left(\frac{w}{z}\right)^{12}$ in simplest form. **2**
- (b) Without using induction, prove that $a^3 + 2a$ is divisible by 3 when a is a positive integer. **3**
- (c) If $x, y > 0$ and $x > y$, prove that $x^4 + y^4 > x^3y + xy^3$. **3**

Question 13 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Prove the following statement, in which n is a positive integer, by using the contrapositive.
- If n leaves a remainder of either 2 or 3 after being divided by 4, then n is not a perfect square. **3**
- (b) (i) Show that $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$. **1**
- (ii) Show by substitution that $\theta = -i \ln(2 + \sqrt{3})$ is a solution to the equation $\cos \theta = 2$. **2**
- (iii) By using the substitution $u = e^{i\theta}$, or otherwise, solve the equation
- $$\frac{e^{i\theta} + e^{-i\theta}}{2} = 2$$
- and hence find another value of θ that solves the equation $\cos \theta = 2$. **2**
- (c) (i) Given that $\frac{d}{dx}(\ln x) = \frac{1}{x}$, show that the Dunstan function, $f(x) = \frac{\ln x}{x}$, is a decreasing function for $x > e$. **2**
- (ii) Using the result in (i), explain why $\frac{\ln a}{a} > \frac{\ln b}{b}$ for $e < a < b$. **1**
- (iii) Rewrite the result in (ii) without logarithms.
- Hence determine if $200^{201} > 201^{200}$ is true or false. **1**

Question 14 (12 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $(x+1)(x^2-x+1) = x^3+1$.

(ii) If $x^2-x+1=0$, find the value of $x^{2022} + \frac{1}{x^{2022}}$.

(b) (i) By expanding $(e^{i\theta} - e^{-i\theta})^3$, show that $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$.

(ii) Hence, solve $8x^3 - 6x + 1 = 0$.

(c) Suppose that x is a real number. Prove that

if $x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 \geq 0$ then $x > 0$.

End of paper

Marks

1

2

3

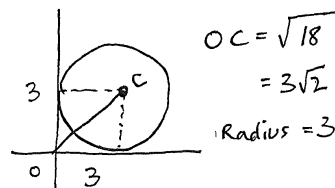
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3

Extension 2 Mathematics 2021 Friday 26/3/21

Q1, The negation of "if p then q "
is " p and not q ". (C)

Q2, $|z - (3+3i)| = 3$



Hence least value of
 $|z| = 3\sqrt{2} - 3$ (A)

Q3, A) False when $y=0$

B) False. $\Delta < 0$

C) False. Consider $x = \frac{1}{2}$

D) True by elimination (D)

Q4, $z^n = \left[\frac{2 \operatorname{cis} \frac{\pi}{6}}{\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)} \right]^n$

$z^n = (\sqrt{2})^n \operatorname{cis} \left(\frac{5n\pi}{12} \right)$

Need $\frac{5n\pi}{12} = \frac{\pi}{2}$

$n = \frac{6}{5}$ (B)

Q5, $z = 2\sqrt{2} e^{i(-\frac{\pi}{4})}$
 $z^{\frac{1}{3}} = \sqrt{2} e^{i(-\frac{\pi}{4} + 2k\pi)}$

$\therefore \theta = \frac{8k\pi - \pi}{12}$

$\theta = \frac{(8k-1)\pi}{12}$ (D)

Q6, $1 + z^2$
 $= 1 + (e^{3i\theta})^2$

$= 1 + e^{i6\theta}$

$= 1 + \cos 6\theta + i \sin 6\theta$

$= 1 + 2\cos^2 3\theta - 1 + 2i \sin 3\theta \cos 3\theta$

$= 2\cos 3\theta (\cos 3\theta + i \sin 3\theta)$

$= 2\cos 3\theta \operatorname{cis} 3\theta$ (B)

Q7, The contrapositive of $p \Rightarrow q$
is $\neg q \Rightarrow \neg p$. (B)

Q8, From $\cos^3 \theta = \frac{1}{8} (e^{i3\theta} + e^{-i3\theta} + 3(e^{i\theta} + e^{-i\theta}))$

$\therefore \cos^3 \theta = \frac{1}{8} (2\cos 3\theta + 3(2\cos \theta))$

$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta$

ie, $\frac{3}{4} \cos \theta = \cos^3 \theta - \frac{1}{4} \cos 3\theta$

(A)

Question 9:

(a) $(x - (2+2i))$ is a factor of $P(x)$

Therefore $(x - (2-2i))$ is also a factor since all the coefficients of $P(x)$ are real. Hence

$$(x - (2+2i))(x - (2-2i)) = (x^2 - 4x + 8) \text{ is a factor.}$$

By inspection, the other factor must be $(x-3)$.

To find the value of a , compare the coefficient of x^2 on both sides of the identity

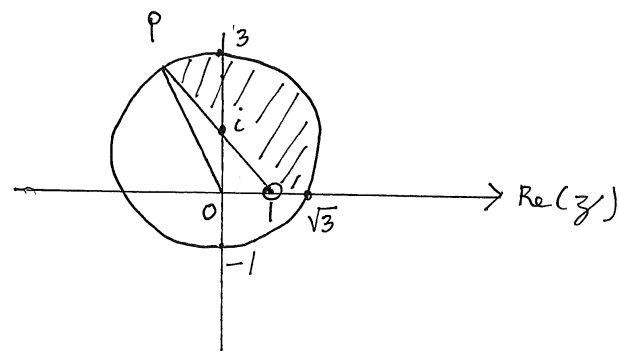
$$(x^2 - 4x + 8)(x-3) \equiv x^3 - ax^2 + 20x - 24.$$

The coefficient of x^2 on the LHS is $(-4-3) = -7$.

Hence $a = 7$.

$$P(x) = (x-3)(x-(2+2i))(x-(2-2i))$$

(b) (i)



(ii) The maximum value of $\text{Arg}(z)$ is $\angle \text{POR}$.

To find P, solve $y = -x+1$ and $x^2 + (y-1)^2 = 4$.

$$\left. \begin{aligned} x^2 + (-x+1-1)^2 &= 4 \\ 2x^2 &= 4 \\ x &= \pm\sqrt{2} \end{aligned} \right\} P = (-\sqrt{2}, \sqrt{2}+1)$$

The gradient of OP is $\frac{\sqrt{2}+1}{-\sqrt{2}}$

$$\tan^{-1}\left(\frac{\sqrt{2}+1}{-\sqrt{2}}\right) = -1.04089$$

$$\angle \text{POR} = -1.04089 + \pi$$

$$= 2.10 \text{ radians}$$

Therefore the maximum value of $\text{Arg}(z)$ is 2.10 rad .

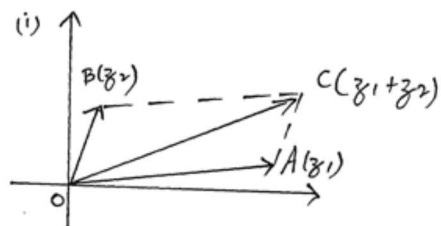
(iii) $\text{Arg}(|z|) = 0$.

Now the $|z|$ measures the distance of z from the origin. That will be a positive real number, lying on the real axis to the right of the origin.

And the argument of any such number is zero.

Question 10:

(a) (i)



$$z_2 = \frac{1}{2} \times \text{cis } \frac{\pi}{3} \times z_1$$

$$z_2 = \frac{1}{2} e^{i\frac{\pi}{3}} z_1$$

$$\begin{aligned} \text{(ii)} \quad \vec{OC} &= \vec{OA} + \vec{OB} \\ &= z_1 + \frac{1}{2} e^{i\frac{\pi}{3}} z_1 \\ &= z_1 \left(1 + \frac{1}{2} e^{i\frac{\pi}{3}}\right) \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \text{The midpoint of } \vec{BC} &= \vec{OB} + \frac{1}{2} \times \vec{OA} \\ &= \frac{1}{2} e^{i\frac{\pi}{3}} z_1 + \frac{1}{2} z_1 \\ &= \frac{1}{2} z_1 (e^{i\frac{\pi}{3}} + 1) \end{aligned}$$

(b) Proof by Contradiction:

Suppose that there are integers m and n such that

$$3m + 21n = 136$$

$$\text{Then } 3(m + 7n) = 3 \times 45 + 1$$

contradiction!

Notice that the LHS is divisible by 3, but the RHS is not divisible by 3.

Therefore the assumption was false.

Therefore there are no integers m and n such that $3m + 21n = 136$.

(c) The contrapositive is:

"if x is odd and y is even then $x^2(y+3)$ is odd".

Let $x = 2k+1$ and $y = 2m$ then

$$\begin{aligned} x^2(y+3) &= (2k+1)^2(2m+3) \\ &= (4k^2 + 4k + 1)(2m + 3) \\ &= 8k^2m + 8km + 2m + 12k^2 + 12k + 3 \\ &= 2[4k^2m + 4km + m + 6k^2 + 6k + 1] + 1 \end{aligned}$$

which is odd.

So if x is odd and y is even, then $x^2(y+3)$ is odd.

Therefore if $x^2(y+3)$ is even, then x is even

or y is odd.

(d) (i)

$$(\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$a - 2\sqrt{ab} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

(ii) From (i) it follows that

$$\frac{xy + yz}{2} \geq \sqrt{xy^2z} \quad \text{--- (1)}$$

$$\text{and } \frac{yz + zx}{2} \geq \sqrt{yz^2x} \quad \text{--- (2)}$$

$$\text{and } \frac{zx + xy}{2} \geq \sqrt{zx^2y} \quad \text{--- (3)}$$

Adding (1) + (2) + (3) gives

$$\frac{xy + yz}{2} + \frac{yz + zx}{2} + \frac{zx + xy}{2} \geq y\sqrt{xz} + z\sqrt{xy} + x\sqrt{yz}$$

$$\frac{2(xy + yz + zx)}{2} \geq y\sqrt{xz} + z\sqrt{xy} + x\sqrt{yz}$$

$$\therefore xy + yz + zx \geq y\sqrt{xz} + z\sqrt{xy} + x\sqrt{yz}$$

Question 11:

(a) From $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

substitute $x = i\theta$ to get

$$e^{i\theta} = 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \dots$$

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} \dots$$

$$= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} \dots\right)$$

But from

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

we get

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots$$

and from

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

we get

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

Hence from

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right)$$

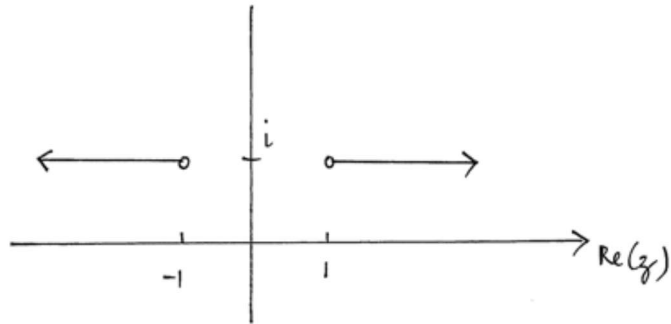
$$\therefore e^{i\theta} = \cos \theta + i \sin \theta$$

as required.

(b)

$$\operatorname{Arg} \left(\frac{z-1-i}{z+1-i} \right) = 0$$

$$\text{i.e., } \operatorname{Arg}(z-(1+i)) - \operatorname{Arg}(z-(-1+i)) = 0$$



$$\begin{aligned} (c) \quad z^4 &= 8\sqrt{2} - 8\sqrt{2}i \\ &= 16 \operatorname{cis} \left(-\frac{\pi}{4} \right) \\ &= 16 e^{i(2k\pi - \frac{\pi}{4})} \\ \therefore z &= 2 e^{i \left(\frac{2k\pi - \frac{\pi}{4}}{4} \right)} \end{aligned}$$

$$z = 2 e^{i \left(\frac{8k-1}{16} \right) \pi}$$

$$\text{when } k = -1, \quad z_1 = 2 e^{-i \frac{9\pi}{16}}$$

$$\text{when } k = 0, \quad z_2 = 2 e^{-i \frac{\pi}{16}}$$

$$\text{when } k = 1, \quad z_3 = 2 e^{i \frac{7\pi}{16}}$$

$$\text{when } k = 2, \quad z_4 = 2 e^{-i \left(\frac{17\pi}{16} \right)} = 2 e^{i \left(\frac{15\pi}{16} \right)}$$

(d)

$$\text{RTP that } |(1+i)z^3 + iz| < \frac{3}{4}, \text{ given } |z| < \frac{1}{2}.$$

A triangle inequality states that

$$|a+b| \leq |a| + |b|.$$

$$\begin{aligned} \text{Hence } |(1+i)z^3 + iz| &\leq |(1+i)z^3| + |iz| \\ &= |(1+i)| \times |z|^3 + |iz| \\ &= \sqrt{2} \times |z|^3 + |iz| \\ &< \sqrt{2} \left(\frac{1}{2} \right)^3 + \frac{1}{2} \\ &= \frac{\sqrt{2} + 4}{8} \\ &< \frac{2+4}{8} \\ &= \frac{3}{4} \end{aligned}$$

$$\therefore |(1+i)z^3 + iz| < \frac{3}{4}$$

Question 12:

$$(a) z = \text{cis} \left(\frac{2\pi}{3} \right)$$

$$w = \sqrt{3} \text{cis} \left(\frac{\pi}{3} \right)$$

$$(i) \frac{w}{z} = \frac{\sqrt{3} \text{cis} \left(\frac{\pi}{3} \right)}{\text{cis} \left(\frac{2\pi}{3} \right)}$$

$$\frac{w}{z} = \sqrt{3} \text{cis} \left(-\frac{\pi}{3} \right)$$

(ii) When $n = 0$,

$$\text{LHS} = (\cos \theta + i \sin \theta)^0 = 1$$

$$\text{RHS} = \cos(0\theta) + i \sin(0\theta) = 1$$

Assume that

$$(\cos \theta + i \sin \theta)^k = \cos(k\theta) + i \sin(k\theta)$$

and try to prove that

$$(\cos \theta + i \sin \theta)^{k+1} = \cos(k+1)\theta + i \sin(k+1)\theta.$$

$$\text{Now LHS} = (\cos \theta + i \sin \theta)^{k+1}$$

$$= (\cos \theta + i \sin \theta)^k \times (\cos \theta + i \sin \theta)$$

$$= (\cos(k\theta) + i \sin(k\theta)) \times (\cos \theta + i \sin \theta) \text{ by the assumption}$$

$$= \cos(k\theta)\cos \theta - \sin(k\theta)\sin \theta + i[\sin(k\theta)\cos \theta + \cos(k\theta)\sin \theta]$$

$$= \cos(k\theta + \theta) + i \sin(k\theta + \theta)$$

$$= \cos(k+1)\theta + i \sin(k+1)\theta$$

$$= \text{RHS}$$

The statement is true by mathematical induction.

$$(iii) \left(\frac{w}{z} \right)^{12}$$

$$= (\sqrt{3} \text{cis} \left(-\frac{\pi}{3} \right))^{12}$$

$$= 3^6 \text{cis}(-4\pi)$$

$$= 3^6$$

$$= 729$$

(b) Now $a^3 + 2a$

$$= a^3 - a + 3a$$

$$= a(a^2 - 1) + 3a$$

$$= a(a-1)(a+1) + 3a$$

Note that this expression is the product of three consecutive integers.

It therefore has a factor of 3.

$$= 3[j] + 3a$$

$$= 3[a+j], \text{ which is divisible by 3.}$$

Hence $a^3 + 2a$ is divisible by 3 for a an positive integer.

(c) RTP that

$$x^4 + y^4 > x^3y + xy^3$$

Consider LHS - RHS

$$= x^4 + y^4 - x^3y - xy^3$$

$$= x^4 - x^3y + y^4 - xy^3$$

$$= x^3(x-y) + y^3(y-x)$$

$$= x^3(x-y) - y^3(x-y)$$

$$= (x^3 - y^3)(x-y)$$

but $x > y > 0$ so
we have the product of
two non-negative terms.

Hence LHS - RHS > 0

$$\therefore x^4 + y^4 > x^3y + xy^3 \quad \#$$

Question 13.

(a) Proof by contrapositive.

The contrapositive is "if n is a perfect square, then n leaves a remainder of either 0 or 1 upon division by 4".

$$\text{Suppose } n = (2j)^2$$

then $n = 4j^2$ and so will leave a remainder of 0 upon division by 4.

$$\text{Suppose } n = (2j+1)^2$$

$$\text{then } n = 4j^2 + 4j + 1$$

$$n = 4(j^2 + j) + 1$$

$n = 4m + 1$ and so will leave a remainder of 1 upon division by 4.

So, if n is a perfect square, then n leaves a remainder of either 0 or 1 upon division by 4.

Hence if n leaves a remainder of either 2 or 3 upon division by 4, then n is not a perfect square.

$$(b) (i) \text{ Now } e^{i\theta} = \cos \theta + i \sin \theta \quad \text{--- (1)}$$

$$e^{-i\theta} = \cos(-\theta) + i \sin(-\theta)$$

$$= \cos \theta - i \sin \theta \quad \text{--- (2)}$$

$$\text{(1) + (2) gives } e^{i\theta} + e^{-i\theta} = 2 \cos \theta$$

$$\text{So } \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

as required.

(b) (ii)

$$\text{Now } \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{--- (1)}$$

substitute $\theta = -i \ln(2+\sqrt{3})$ into (1) to get

$$\cos \theta = \frac{e^{-i(i \ln(2+\sqrt{3}))} + e^{i(i \ln(2+\sqrt{3}))}}{2}$$

$$= \frac{e^{\ln(2+\sqrt{3})} + e^{\ln(2+\sqrt{3})^{-1}}}{2}$$

$$= \frac{\{(2+\sqrt{3}) + (2+\sqrt{3})^{-1}\}}{2}$$

$$= \frac{\left\{2+\sqrt{3} + \frac{2-\sqrt{3}}{4-3}\right\}}{2}$$

$$= \frac{\{2+\sqrt{3} + 2-\sqrt{3}\}}{2}$$

$$= 2$$

Hence $\theta = -i \ln(2+\sqrt{3})$ is a solution to $\cos \theta = 2$.

(iii) To solve $\frac{e^{i\theta} + e^{-i\theta}}{2} = 2$ sub $u = e^{i\theta}$ to get

$$\frac{u + \frac{1}{u}}{2} = 2$$

$$u^2 - 4u + 1 = 0$$

$$u = \frac{4 \pm \sqrt{12}}{2}$$

$$u = 2 \pm \sqrt{3}$$

$$\therefore e^{i\theta} = 2 \pm \sqrt{3}$$

$$i\theta = \ln(2 \pm \sqrt{3})$$

$$\therefore \theta = \frac{1}{i} \ln(2 \pm \sqrt{3})$$

$$\therefore \theta = -i \ln(2 \pm \sqrt{3})$$



$$\theta = -i \ln(2+\sqrt{3})$$

$$\theta = -i \ln(2-\sqrt{3})$$

$\therefore$ Another solution is $\theta = -i \ln(2-\sqrt{3})$

c) (i) $f(x) = \frac{\ln x}{x}$

$$f'(x) = \frac{x(\frac{1}{x}) - \ln x}{x^2}$$

$$f'(x) = \frac{1 - \ln x}{x^2}$$

The denominator is always positive.

The numerator is negative if $x > e$.

Hence $f(x)$ is a decreasing function for $x > e$.

(iii) Now $\frac{\ln a}{a} > \frac{\ln b}{b}$

$$b \ln a > a \ln b$$

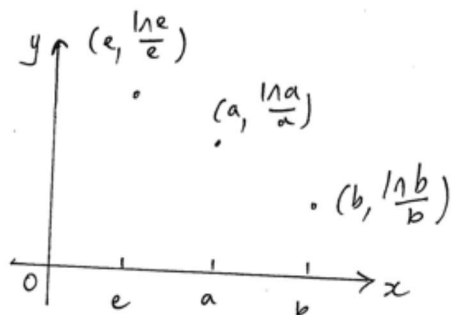
$$\ln a^b > \ln b^a$$

$$\therefore a^b > b^a \quad \text{--- (2)}$$

when $e < a < b$.

Hence $200^{201} > 201^{200}$ is true using the result called (2).

(ii) Graphically, we have



From this graph, clearly

$$\frac{\ln a}{a} > \frac{\ln b}{b}$$

for $e < a < b$.

Question 14:

(a) (i) LHS = $(x+1)(x^2-x+1)$
 $= x^3 - x^2 + x + x^2 - x + 1$
 $= x^3 + 1$

Hence $(x+1)(x^2-x+1) = x^3+1$, as required

(ii) If $x^2-x+1=0$ then $x^3+1=0$.

So x is either -1 or a complex cube root of -1 .

But $x = -1$ is not a solution of $x^2-x+1=0$, so

x must be a cube root of -1 .

Hence $x^{2022} + \frac{1}{x^{2022}}$
 $= (x^3)^{674} + \frac{1}{(x^3)^{674}}$
 $= (-1)^{674} + \frac{1}{(-1)^{674}}$
 $= 1 + \frac{1}{1}$
 $= 2.$

(b) (i)

$$\begin{aligned}(e^{i\alpha} - e^{-i\alpha})^3 &= (e^{i\alpha})^3 - 3(e^{i\alpha})^2(e^{-i\alpha}) + 3(e^{i\alpha})(e^{-i\alpha})^2 - (e^{-i\alpha})^3 \\&= e^{i3\alpha} - 3e^{i\alpha} + 3e^{-i\alpha} - e^{-i3\alpha} \\&= e^{i3\alpha} - e^{-i3\alpha} - 3(e^{i\alpha} - e^{-i\alpha}) \\&= 2i \sin 3\alpha - 3(2i \sin \alpha) \\&= 2i \sin 3\alpha - 6i \sin \alpha\end{aligned}$$

$$\begin{aligned}\text{but } (e^{i\alpha} - e^{-i\alpha})^3 &= (2i \sin \alpha)^3 \\&= -8i \sin^3 \alpha\end{aligned}$$

$$\text{so } -8i \sin^3 \alpha = 2i \sin 3\alpha - 6i \sin \alpha$$

$$4 \sin^3 \alpha = 3 \sin \alpha - \sin 3\alpha$$

$$\therefore \sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha, \text{ as required}$$

(ii) Now $8x^3 - 6x + 1 = 0$

$$1 = 6x - 8x^3$$

$$\frac{1}{2} = 3x - 4x^3$$

let $x = \sin \alpha$ and solve $\sin 3\alpha = \frac{1}{2}$

$$3\alpha = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \dots$$

$$\alpha = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \dots$$

$$\therefore x = \sin\left(\frac{\pi}{18}\right), \sin\left(\frac{5\pi}{18}\right), \sin\left(\frac{25\pi}{18}\right).$$

(c) RTP that

$$\text{if } x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 \geq 0 \text{ then } x > 0.$$

Proof by contrapositive:

$$\text{Suppose } x \leq 0.$$

Consider

$$x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4$$

$$\begin{aligned}&= (x^5 + 3x^3 + 3x) - (4x^4 + x^2 + 4) \\&\quad \uparrow \quad \uparrow \quad \uparrow \quad \quad \uparrow \quad \uparrow \quad \uparrow \\&\quad (-) \quad (-) \quad (-) \quad \quad (+) \quad (+) \quad (+)\end{aligned}$$

The expression above is a negative (or zero) number minus a positive number.

That will definitely give a negative result.

$$\text{Hence } (x^5 + 3x^3 + 3x) - (4x^4 + x^2 + 4) < 0$$

$$\text{So if } x < 0 \text{ then } x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 < 0.$$

So therefore if

$$x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 \geq 0,$$

$$\text{then } x > 0.$$